谭博予

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教育背景

复旦大学　复旦管院 **2017.09-2021.07**
理学学士　统计学 **上海，中国**

- 绩点：3.75/4.00，排名：7/216(院)、5/54(系)
- 主要课程：数据挖掘、数据库与企业数据管理、最优化方法、运筹学、博弈论、属性数据分析、多元分析、抽样调查、随机过程、回归分析、时间序列分析、统计推断、概率论与数理统计、数学分析、线性代数

复旦大学　复旦管院 **2021.09-2023.07**
理学硕士　数据科学与商业分析（应用统计） **上海，中国**

- 绩点：3.67/4.00
- 主要课程：分布式计算、深度学习、机器学习、数据可视化、统计计算、极值理论、决策与优化

工作经历

- **Paypal-全球数据科学与分析部-Venmo风控模型团队　　机器学习算法岗　　2023.7-至今**

1. Venmo AddFunds 2024 models：余额充值风控模型

- 针对Venmo Standard/Instant Add funds产品建立实时风控OMOE深度学习模型，单任务建模抓取盗卡/盗银行账户的损失。
- 利用Reject Inference给部分被拒绝的交易打上标签后加入训练集以提高模型风控表现，进行变量选择后通过Multi-segment WOE transformation进行变量再选择和数据预处理后，训练Tensorflow OMOE深度学习模型。
- 相对于benchmark models，新模型的Catch Rate分别能在Standard上从57.9%提高到78.9% @OP4%，在Instant上从28.5%提高到50.0% @OP2.3%。模型上线后第一阶段能够在同样损失预算下，对Standard add funds flow降低25.7%的Action Rate，对应于增加每年$169.2MM的TPV；对Instant add funds flow降低39.4%的Action Rate，对应增加每年$698.4MM的TPV。

2. Venmo AddFI 2024 models：加卡/加银行账户风控模型

- 针对Venmo AddCard/AddBank产品建立实时风控MMOE深度学习模型，双任务建模抓取坏AddFI和脚本攻击带来的损失。
- 重新整理Venmo AddFI的tagging逻辑，全面覆盖盗卡/坏卡/坏账户和脚本攻击等情况。新创建并采用大量新特征，新模型中Top500变量中有超过40%是新变量，包括Profile IP_State和Card_State Match实时变量，ADDFI velocity实时变量和针对FI level的离线刷新变量等。
- 相对于benchmark models，新模型的Catch Rate在AddBank flow上从39.1%提高到73.8% @OP8.5%，在AddCard上从35%提高到68% @OP10%。根据测试集上的估计，在可接受FPR下新模型能够额外带来每年$7.3MM net loss saving。

3. Venmo ATO Models：负责盗号风控相关模型

- 负责Venmo P2P ATO(Account Take Over)的Rapid Model Refresh实时模型，从Control Group上看该flow ATO loss income pressure为$60MM每年。模型每两个月自动刷新一次，在0.45%的Action Rate下能够抓取96.36%的Venmo P2P ATO loss。
- 只根据设备信息和账户id设计新特征并训练轻量模型，便于部署到新的风控节点和防范新产品上线的ATO损失。

4. XAI models：建立可解释机器学习模型以修正或解释原风控模型

- 自动化决策：对原风控模型建立其对应的XAI模型，根据Shap Value和Reason Code进行再分桶以强化原始模型表现。
- 输出可解释性：在现有模型Real-time Issuance Buyer Abuse Model基础上建立其对应的XAI Model，输出最影响模型得分的具有指向性的Top 5变量，将变量的含义转换后集成到SCM UI里加速人工看Case的速度。模型上线后能降低20%的Case Review时间。

5. Patents Approved for Filling

- (OCP.D2024.105668.US1) Dynamically Adjustable eXplainable Artificial Intelligence (XAI) Model
- (OCP.D2024.105077.WO1) DATA ANNOTATION SYSTEM FOR TRAINING ARTIFICIAL INTELLIGENCE MODELS USING DATA SETS WITH EXCLUDED DATA POPULATIONS

荣誉及技能兴趣

- 获得荣誉：复旦大学优秀毕业生、复旦大学优秀学生一等奖学金、2020年全国大学生数学建模国家级二等奖、2020年上海新能源汽车大数据竞赛二等奖
- 编程语言：熟练使用Python、SQL、R语言、Tensorflow和Pytorch深度学习框架，熟悉Hadoop，Spark和Linux Shell开发环境
- 语言能力：IELTS6.5、CET-6
- 兴趣爱好：热爱健身，游泳和骑行

Bruce Tan
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EDUCATION

Fudan University, School of Management **Sep.2017-Jun.2021**
Bachelor's Degree, major in Statistics **Shanghai, China**

- GPA: 3.75/4.00, Ranking: 7/216 (3%)
- Main Courses: Data Mining, Database, Optimization, Operations Research, Game Theory, Categorical Data Analysis, Multivariate Analysis, Random Process, Regression Analysis, Time Series Analysis, Statistical Inference, Probability Theory and Mathematical Statistics, Mathematical Analysis, Linear Algebra

Fudan University, School of Management **Sep.2021-Jun.2023**
Master's Degree, major in Data Science (Applied Statistics) **Shanghai, China**

- GPA: 3.67/4.00
- Main Courses: Distributed computing, DL, ML, Data Visualization, Statistical Computing, EVT, Optimizing

WORKING EXPERIENCE

Paypal - Global Analytics and Data Science - Machine learning Scientist **Jul.2023-Now**

1. Venmo AddFunds 2024 models

- Build single-task online risk models of OMOE structure for Venmo standard and instant add funds products to capture losses from stolen cards/bank accounts.
- Use Reject Inference to label some rejected transactions into training dataset to improve the model's performance, train the Tensorflow OMOE model after feature selection and multi-segment WOE transformation for vaiables second selection and data preprocessing.
- Compared with the baseline models, the catch rate of new model can be increased from 57.9% to 78.9% @OP4% on standard flow and from 28.5% to 50.0% @OP2.3% on Instant flow. In the first phase after the models are online, under the same loss budget, the Action Rate for Standard add funds was reduced by 25.7%, corresponding to an increase in TPV of $169.2MM per year; the Action Rate for Instant add funds was reduced by 39.4%, corresponding to an increase in TPV of $698.4MM per year.

2. Venmo AddFI 2024 models

- Build multi-task online risk models of MMOE structure Venmo Add Card and Add Bank products to capture losses caused by bad AddFI and script attacks.
- Refined the tagging logic of Venmo AddFI to fully cover stolen cards/bad cards/bad accounts and script attacks. A large number of new features are newly created and adopted. More than 40% of the Top500 features in the new models are newly created, including Profile IP_State and Card_State Match real-time variables, ADDFI velocity real-time variables, and offline batch features of FI level.
- Compared to the baseline model, the capture rate of the new model increased from 39.1% to 73.8% @OP8.5% on AddBank flow and from 35% to 68% @OP10% on AddCard. According to the estimation on the test set, the new models can bring an additional net loss savings of $7.3MM per year under actionable FPR.

3. Venmo ATO related models

- Responsible for the Rapid Model Refresh online model of Venmo P2P ATO (Account Take Over). From the Control Group, the ATO loss income pressure of this flow is about $60MM. The model is automatically refreshed every two months and can capture 96.36% of Venmo P2P ATO loss at an Action Rate of 0.45%.
- New features are designed and lightweight models are trained based only on device information and account IDs, making it easier to deploy them to new risk control nodes and prevent ATO losses when new products are launched.

4. XAI models:

- Automated decision-making: Establish a corresponding XAI model for the original risk control model, and re-bucket it according to Shap Value and Reason Code to enhance the performance of the original model.
- Output interpretability: Based on the existing Risk Model Real-time Issuance Buyer Abuse Model, established a corresponding XAI Model, output the top 5 indicative variables that have the greatest impact on the model score, and converted the meaning of the variables to the SCM UI to speed up manual case review. After the model is launched, it can reduce the average case review time by 20%.

5. Patents Approved for Filling

- (OCP.D2024.105668.US1) Dynamically Adjustable eXplainable Artificial Intelligence (XAI) Model
- (OCP.D2024.105077.WO1) DATA ANNOTATION SYSTEM FOR TRAINING ARTIFICIAL INTELLIGENCE MODELS USING DATA SETS WITH EXCLUDED DATA POPULATIONS

AWARDS & SKILLS

- AWARDS: Outstanding Graduates of Fudan University, First Class Scholarship for Outstanding Students of Fudan University, 2020 National University Student Mathematical Modeling National Second Prize
- SKILLS: Proficient in using Python, SQL, R language, Tensorflow, and Pytorch deep learning frameworks, familiar with Hadoop, Spark, and Linux Shell development environments
- LANGUAGE: IELTS6.5, CET-6
- INTERESTS: Love Gym, Swimming and Cycling